

Citation Edits & Reference Guide

A line-by-line guide to addressing the remaining 'Cite' comments

1. Introduction (Mission Context & Tools)

Comment [2]

Location:

...goals of the upcoming Ariel [CITE] mission...

Insert the citation for the official Ariel mission whitepaper.

Tinetti, G., Eccleston, P., Haswell, C., et al. (2018). A chemical survey of exoplanets with ARIEL. Experimental Astronomy, 46(1), 135-209. <https://doi.org/10.1007/s10686-018-9598-x>

Comment [4]

Location:

Existing [CITE] searches for habitable exoplanets have included...

Add a citation recognizing historical transmission spectroscopy surveys.

de Kok, R. J., Brogi, M., Snellen, I. A. G., et al. (2013). Detection of water vapour and constraints on carbon monoxide and methane in the atmosphere of HD 209458b. Astronomy & Astrophysics, 554, A82.

Comment [11]

Location:

To do so, we use the multirex library [CITE]...

Cite the Alfnor paper, which is the underlying framework that powers the multirex package.

Changeat, Q., & Al-Refaie, A. F. (2020). Alfnor: A retrieval-based population study of Ariel targets. The Astronomical Journal, 160(2), 80.

Comment [12]

Location:

...that leverages the TauREx 3 radiative transfer code [CITE]...

Cite the TauREx 3 framework. (Note: Comment [17] also asks for this exact citation).

Al-Refaie, A. F., Changeat, Q., Waldmann, I. P., & Tinetti, G. (2021). TauREx 3: A fast, dynamic, and extendable framework for retrievals. The Astrophysical Journal, 917(1), 37.

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Comment [22]

Location:

The dataset used in this study was drawn from the Ariel Big Challenge (Tier 2) [CITE]...

Cite the Ariel Data Challenge paper which establishes this specific dataset format.

Yip, K. H., Changeat, Q., Nikolaou, N., et al. (2021). Peeking inside the black box: Interpreting deep-learning models for exoplanet atmospheric retrievals. The Astronomical Journal, 162(5), 195.

2. Methodology (Physics & Data Generation)

Comment [20] & [21]

Location:

After the description of TauREx 3 limitations (1D assumptions, etc.).

Add citations to back up the physical data sources (ExoMol) and 3D limitations.

Tennyson, J., Yurchenko, S. N., Al-Refaie, A. F., et al. (2020). The 2020 release of the ExoMol database: Molecular line lists for exoplanet and other hot atmospheres. Journal of Quantitative Spectroscopy and Radiative Transfer, 255, 107228.

Caldas, A., Leconte, J., Selsis, F., et al. (2019). Effects of a fully 3D atmospheric structure on exoplanet transmission spectra. Astronomy & Astrophysics, 623, A161.

Comment [25] through [31]

Location:

Table I. Parameter grid choices for synthetic spectra...

Update the caption to read: '...following the simulated Ariel data parameters described by Yip et al. (2021).'

Yip, K. H., Changeat, Q., Nikolaou, N., et al. (2021). Peeking inside the black box: Interpreting deep-learning models for exoplanet atmospheric retrievals. The Astronomical Journal, 162(5), 195.

Comment [54]

Location:

...with a constant signal-to-noise ratio (SNR) of 15 [CITE] applied...

Cite the source that establishes SNR=15 as a viable threshold for this type of detection.

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Duque-Castaño, D. S., et al. (2024). Machine-assisted classification of potential biosignatures in earth-like exoplanets using low signal-to-noise ratio transmission spectra. *arXiv preprint arXiv:2407.19167v2*.

3. Machine Learning Architectures

Comment [35]

Location:

This approach was selected for its ability to model complex, non-linear relationships... near the detection thresholds. [CITE]

Cite the creators of XGBoost.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785-794).

Comment [40]

Location:

Batch normalization and dropout ($p = 0.4$) were applied after each layer to improve training stability and reduce overfitting. [CITE]

Cite the standard Deep Learning textbook by Goodfellow et al. to justify these regularization methods.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press. <http://www.deeplearningbook.org>